

ECO387C - Fall 2024

Problem Set 2 Solutions

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Question 1

a Setting up the Lagrangian for each agent $\ell \in \{A, B\}$:

$$\begin{aligned}\mathcal{L}^\ell = \log(c_0^\ell) + \beta^\ell \left[p \cdot \log(c_1^\ell) + (1-p) \cdot \log(c_2^\ell) \right] &+ \lambda_0^\ell \left[e_0^\ell - c_0^\ell - b^\ell \right] \\ &+ \lambda_1^\ell \left[e_1^\ell + (1+r)b^\ell - c_1^\ell \right] \\ &+ \lambda_2^\ell \left[e_2^\ell + (1+r)b^\ell - c_2^\ell \right]\end{aligned}$$

Or using bond price q instead of interest rate:

$$\begin{aligned}\mathcal{L}^\ell = \log(c_0^\ell) + \beta^\ell \left[p \cdot \log(c_1^\ell) + (1-p) \cdot \log(c_2^\ell) \right] &+ \lambda_0^\ell \left[e_0^\ell - c_0^\ell - qb^\ell \right] \\ &+ \lambda_1^\ell \left[e_1^\ell + b^\ell - c_1^\ell \right] \\ &+ \lambda_2^\ell \left[e_2^\ell + b^\ell - c_2^\ell \right]\end{aligned}$$

b The first order conditions of this problem are:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial c_0^\ell} = 0 &\implies \frac{1}{c_0^\ell} = \lambda_0^\ell \\ \frac{\partial \mathcal{L}}{\partial c_1^\ell} = 0 &\implies \frac{\beta^\ell p}{c_1^\ell} = \lambda_1^\ell \\ \frac{\partial \mathcal{L}}{\partial c_2^\ell} = 0 &\implies \frac{\beta^\ell (1-p)}{c_2^\ell} = \lambda_2^\ell \\ \frac{\partial \mathcal{L}}{\partial b^\ell} = 0 &\implies \lambda_0^\ell = (1+r)(\lambda_1^\ell + \lambda_2^\ell)\end{aligned}$$

Hence, agent ℓ 's consumption-Euler equation is:

$$\frac{1}{c_0^\ell} = \beta^\ell (1+r) \left[\frac{p}{c_1^\ell} + \frac{1-p}{c_2^\ell} \right]$$

c The definition of equilibrium in this economy is as follows

Definition 1 (Competitive Equilibrium) A competitive equilibrium is a set of price $\{r\}$ and a set of allocations $\{c_0^\ell, c_1^\ell, c_2^\ell, b^\ell\}$ for $\ell \in \{A, B\}$ such that

- (a) *Individual optimality: Taking price r as given, the allocations solve each household's problem*
- (b) *Market clearing: The market for goods clears in each period and state*

$$c_\tau^A + c_\tau^B = e_\tau^A + e_\tau^B \quad \text{for each } \tau = 0, 1, 2$$

and the market for assets (bonds) clears at $\tau = 0$.

$$b^A + b^B = 0$$

d Define the price of the bond as $q = (1 + r)^{-1}$. Under the given numbers, combining the Euler equation with the budget constraint for each agent gives

- Agent A:

$$q = \frac{\frac{1}{2} \cdot \frac{1}{2} \cdot (\frac{19}{8} - q \cdot b^A)}{b^A + 1} + \frac{\frac{1}{2} \cdot \frac{1}{2} (\frac{19}{8} - q \cdot b^A)}{b^A + 3}$$

- Agent B:

$$q = \frac{\frac{1}{3} \cdot \frac{1}{2} \cdot (\frac{21}{8} - q \cdot b^B)}{b^B + 5} + \frac{\frac{1}{3} \cdot \frac{1}{2} (\frac{21}{8} - q \cdot b^B)}{b^B + 3}$$

Rearranging both equations in terms of q and combining the two equations gives:

$$q = \frac{\frac{19}{32} \left(\frac{1}{b^A + 1} + \frac{1}{b^A + 3} \right)}{1 + \frac{b^A}{4} \left(\frac{1}{b^A + 1} + \frac{1}{b^A + 3} \right)} = \frac{\frac{21}{48} \left(\frac{1}{b^B + 5} + \frac{1}{b^B + 3} \right)}{1 + \frac{b^B}{6} \left(\frac{1}{b^B + 5} + \frac{1}{b^B + 3} \right)}$$

Using the bond market clearing condition ($b^B = -b^A$), the equation above results in the following cubic equation.

$$139(b^A)^3 - 422(b^A)^2 - 923b^A + 1206 = 0 \quad \Leftrightarrow \quad (b^A - 1)(139(b^A)^2 - 283b^A - 1206) = 0$$

While real solutions exist to the second quadratic equation, they cannot constitute the equilibrium because they are associated with negative consumption. Hence, the equilibrium bond holdings are $b^A = 1$ and $b^B = -1$. We can use these to back out the following equilibrium allocations and prices:

Using bond price								
Price	Consumption						Bond	
q	c_0^A	c_1^A	c_2^A	c_0^B	c_1^B	c_2^B	b^A	b^B
$\frac{3}{8}$	2	2	4	3	4	2	1	-1

Using interest rate								
Price	Consumption						Bond	
$1+r$	c_0^A	c_1^A	c_2^A	c_0^B	c_1^B	c_2^B	b^A	b^B
$\frac{8}{3}$	2	2	4	3	4	2	$\frac{3}{8}$	$-\frac{3}{8}$

Question 2

Please see the attached MATLAB code for the solutions.

2.1 - Interpolation

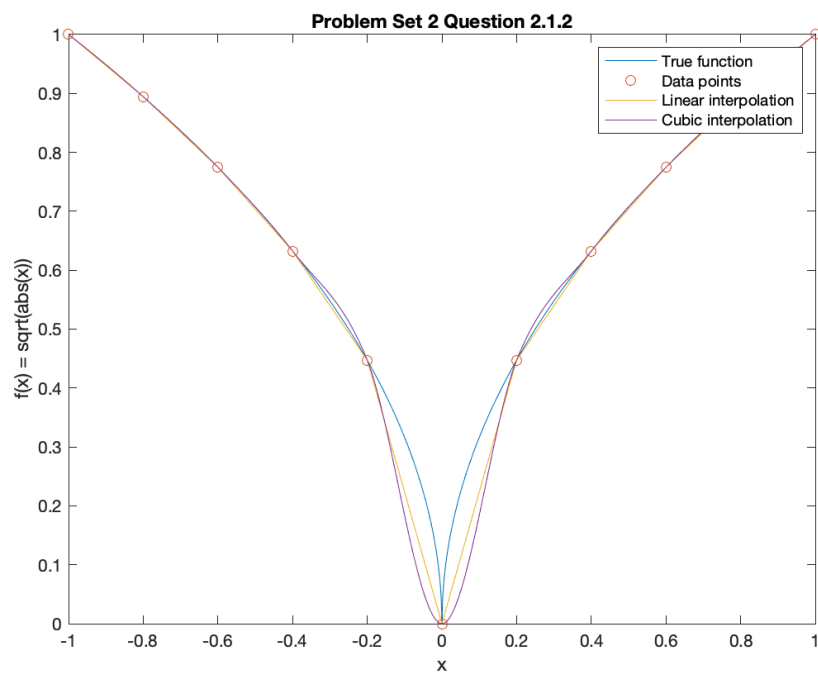
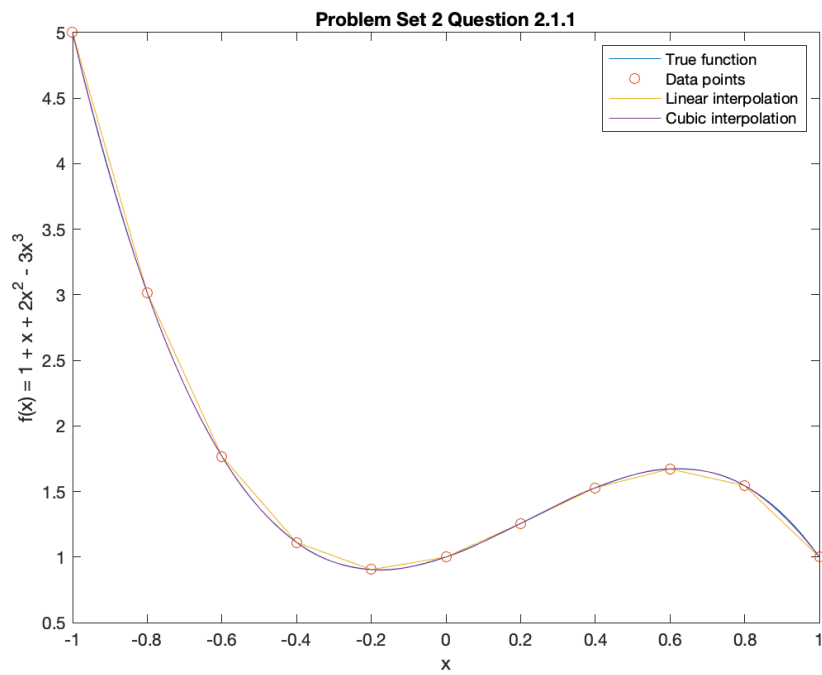
1. In this situation, where the true function is smooth and differentiable everywhere, cubic splines would be a better choice than linear splines since cubic splines allow for curvature while linear splines cannot.
2. If you think that the true function is rather ill-behaved, such as having kink-points and discontinuities, linear splines are the better choice. They do not impose curvature when none exists, which cubic splines can't help but do so.

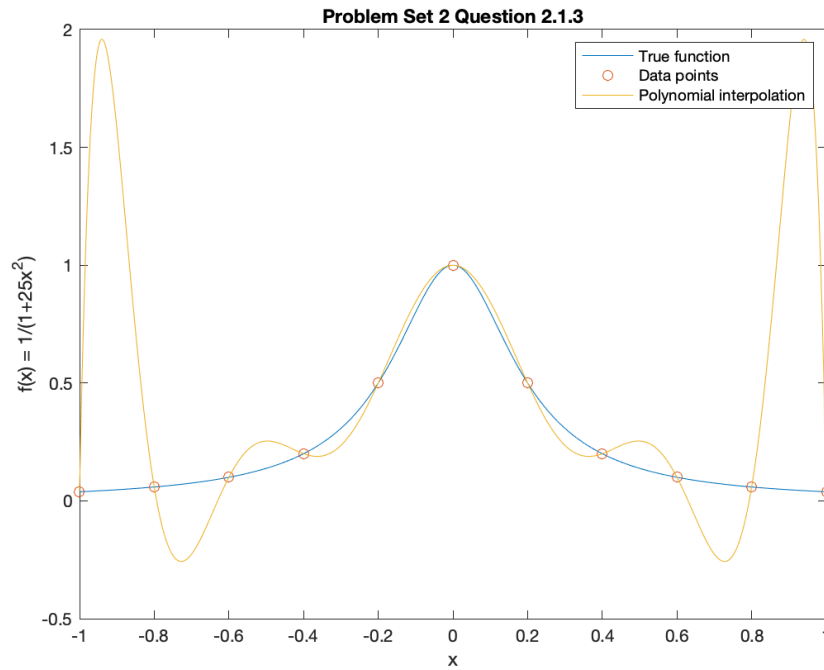
Overall, the 'best' interpolation scheme is heavily context-specific. When approximating a function, it's best to first carefully think about what you think the function should look like. E.g. if there is a discrete choice in your problem, then cubic splines are a poor choice for approximating the value and policy functions.

3. Note the highly oscillatory behaviour at the end-points of the interval. Polynomials can perfectly fit any set of data points, but we don't use them in practice because they prefer certain shapes over others. Not only is there high interpolation error in this scenario, but the **extrapolation error** would also likely be high. For more details, see Runge's phenomenon.

2.2 - Root-Finding

1. $p^* \approx 0.1428$.





2.3 - Numerical Integration

1. The change in consumer surplus can be analytically shown to be an increase by 4.

$$\Delta(\text{CS}) = \int_1^4 2p^{-0.5} dp = 4$$

- 2.

$$\mathbb{E}(\sin(\tilde{X}) + \exp(\tilde{X})) \approx 10.8618$$